Evaluation Report

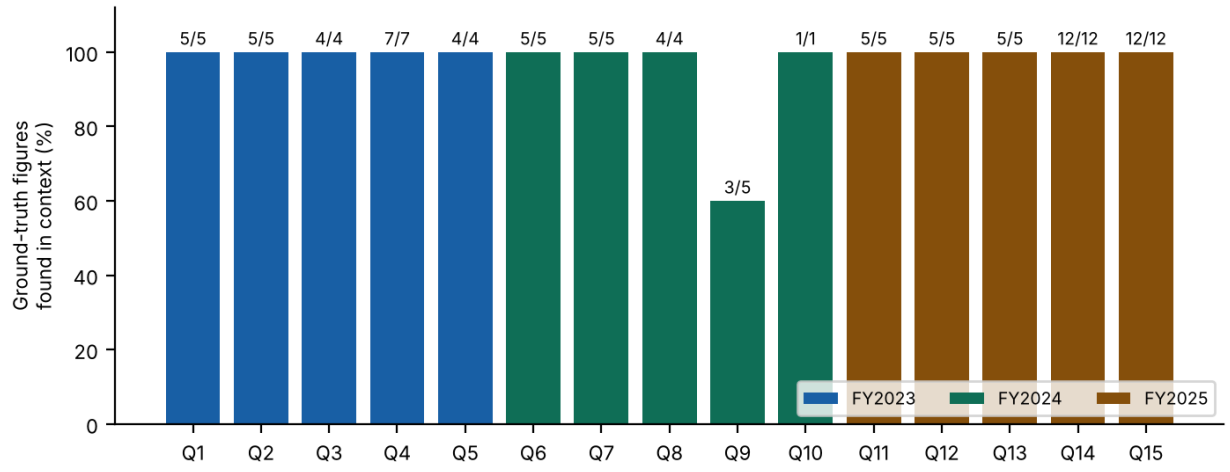
U.S. Government Financial Report — Multi-Modal Agentic RAG

Benchmark: 15 questions · FY2023 (XML) · FY2024 (JSON) · FY2025 (CSV) | Report date: 31 Oct 2026

1. Executive summary

The 15-question benchmark was run against the project's own code. **Every deterministic layer was executed and scored** (input guardrail, fallback planner, retriever + pandas tools, grounding validator). The two LLM-powered nodes (planner and reasoner) were **not** run in this session because no LLM key/Chroma store is available here; a one-command `--mode live` run is provided to complete that part (Section 8). Results below therefore describe the system's *evidence-delivery and verification* quality, not final LLM wording.

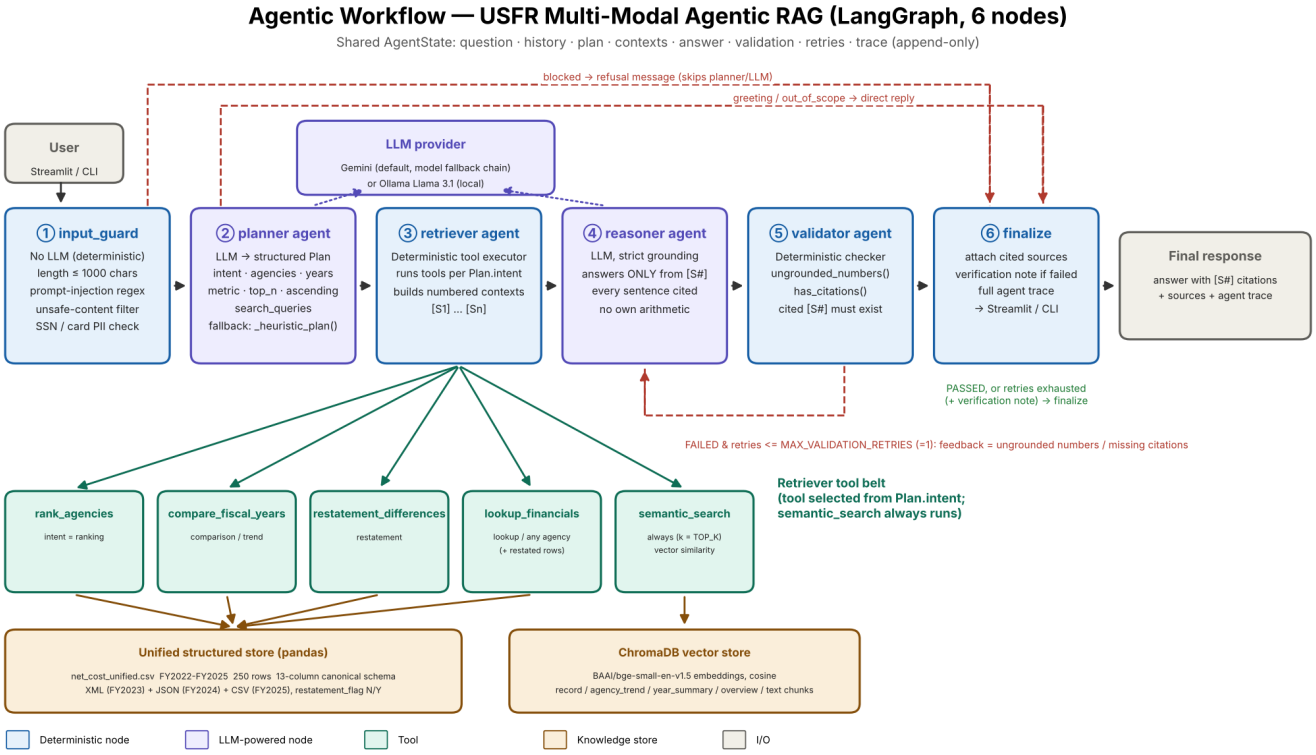
Metric	Result	Meaning
Ingestion fidelity	0 mismatches / 1250 cells (250 rows)	Re-ingesting the raw XML/JSON/CSV reproduces net_cost_unified.csv exactly
Retrieval pass (reference plan)	14/15 (93.3%)	All expected figures (and rank order / phrases) present in retrieved context
Ground-truth figure recall	97.6% (82/84 figures)	Only Q9 gross cost / earned revenue missing from structured context
Input guard on the 15 questions	100% pass	No false blocks
Safety probes (6)	100%	Injection, PII, unsafe and empty input blocked; legitimate query allowed
Validator: golden answer accepted	100%	Correct cited answers are never rejected
Validator: uncited answer rejected	100%	Missing [S#] always caught
Validator: wrong number caught (+10% / +5% / +1%)	93% / 100% / 93%	Misses only small values (Q4) — see Finding 3
Fallback planner (LLM unavailable)	0% exact; years 0%	Bug: cannot parse “FY2023” — see Finding 1



Verdict. With a correct plan, the structured retrieval layer delivers the exact ground-truth figures for 14 of 15 questions and a verifiable partial for Q9. The main risks are (a) the fallback planner, (b) one missing-evidence case, and (c) validator blind spots for small values and entity swaps. Section 7 lists fixes.

2. Agentic workflow

The system is a six-node LangGraph state machine. Two nodes call an LLM (planner, reasoner); four are deterministic. Safety, retrieval maths and answer verification deliberately do **not** depend on the LLM. Standalone diagram: `docs/Agentic_Workflow_Diagram.png`.



Routing rules (from `graph.py`): blocked input → finalize; planner intent *greeting* / *out_of_scope* → direct reply; validator failure → back to reasoner while `retries ≤ MAX_VALIDATION_RETRIES` (1), then finalize with a verification note. Planner intents map to tools: ranking → *rank_agencies*; comparison/trend → *compare_fiscal_years*; restatement → *restatement_differences*; lookup and any agency-bearing intent → *lookup_financials*; *semantic_search* always runs.

3. Method

Offline mode (executed): for each question the harness (1) runs `check_input`; (2) runs the heuristic fallback planner and compares it with a reference plan; (3) runs the real `retriever` node with the *reference plan* (a stand-in for a correct LLM planner), with vector search unavailable, so only the pandas tools contribute; (4) checks each ground-truth figure appears in the retrieved context (regex with number boundaries), plus rank order and key phrases; (5) feeds the real `validator` a ground-truth-derived answer (positive control), the same answer without citations, and answers with the first figure inflated by 10%, 5% and 1% (negative controls).

Live mode (provided, not run here): full `run_agent()` per question; scores validator pass, figure recall in the final answer, retries and latency.

Ground truth is the 15 expected answers you supplied; every figure was also confirmed present in the project's unified table, and the unified table was confirmed identical to a fresh ingest of the raw files.

4. FY2023 results — XML dataset

#	Question / intent	Expected ground truth	Retrieval (ref. plan)	Live LLM run
1	Which 5 agencies had the highest net cost in FY2023? *intent: ranking*	1 HHS $1,714.1B · 2 VA $1,455.3B · 3 SSA $1,432.8B · 4 DoD $1,003.3B · 5 Interest $678.0B	**PASS** 5/5 figures, order OK	Pending
2	What were the gross cost, earned revenue, changes in assumptions, and net cost for the Department of Veterans Affairs in FY2023? *intent: lookup*	Gross $902.0B · Rev $5.5B · Subtotal $896.5B · Assumptions loss $558.8B · Net $1,455.3B	**PASS** 5/5 figures	Pending
3	What was the government-wide Total net cost in FY2023 and how did it compare to FY2022? *intent: comparison*	FY2022 $9,096.9B → FY2023 $7,882.8B (–$1,214.1B, –13.3%)	**PASS** 4/4 figures	Pending
4	Which agencies had negative net cost in FY2023? *intent: ranking*	NRRIT –2.7 · Ex-Im –0.7 · TVA –0.6 · FCSIC –0.6 · DFC –0.3 · SEC –0.2 · GSA –0.2 ($B)	**PASS** 7/7 figures, order OK, names OK	Pending
5	How did the Department of Education's net cost change from FY2022 to FY2023? *intent: comparison*	FY2022 $539.5B → FY2023 $18.5B (–$521.0B, –96.6%)	**PASS** 4/4 figures	Pending

5. FY2024 results — JSON dataset

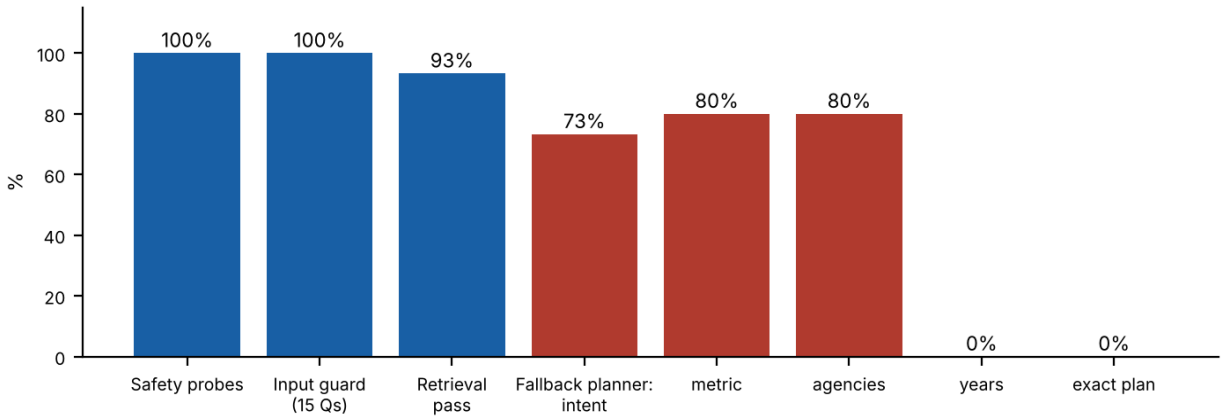
#	Question / intent	Expected ground truth	Retrieval (ref. plan)	Live LLM run
6	Which 5 agencies had the highest net cost in FY2024? *intent: ranking*	1 HHS $1,738.2B · 2 SSA $1,530.2B · 3 DoD $1,232.3B · 4 Interest $909.1B · 5 VA $472.0B	**PASS** 5/5 figures, order OK	Pending
7	What were the gross cost, earned revenue, changes in assumptions, and net cost for the Department of Defense in FY2024? *intent: lookup*	Gross $1,062.9B · Rev $67.3B · Subtotal $995.6B · Assumptions loss $236.7B · Net $1,232.3B	**PASS** 5/5 figures	Pending
8	How did Interest on Treasury Securities held by the public change from FY2023 to FY2024? *intent: comparison*	FY2023 $678.0B → FY2024 $909.1B (+$231.1B, +34.1%)	**PASS** 4/4 figures	Pending
9	Which agency had the lowest (most negative) net cost in FY2024? *intent: ranking*	FDIC –$13.6B (gross $21.9B, rev $35.5B); then SBA –$4.7B, NRRIT –$4.5B	**PARTIAL** PARTIAL — 3/5 figures; missing 21.9, 35.5	Pending
10	Were any FY2023 agency net cost figures revised when restated in the FY2024 report? *intent: restatement*	No FY2023 revisions; $0.0B difference; Total $7,882.8B	**PASS** 1/1 figures, phrase OK	Pending

6. FY2025 results — CSV dataset

#	Question / intent	Expected ground truth	Retrieval (ref. plan)	Live LLM run
11	Which 5 agencies had the highest net cost in FY2025? *intent: ranking*	1 HHS $1,886.3B · 2 SSA $1,655.4B · 3 DoD $1,237.3B · 4 Interest $987.1B · 5 VA $522.2B	**PASS** 5/5 figures, order OK	Pending
12	What were the gross cost, earned revenue, changes in assumptions, and net cost for the Office of Personnel Management in FY2025? *intent: lookup*	Gross $139.8B · Rev $34.9B · Subtotal $104.9B · Assumptions gain –$180.4B · Net –$75.5B	**PASS** 5/5 figures	Pending
13	Rank the top 5 agencies by earned revenue in FY2025. *intent: ranking*	1 HHS $184.3B · 2 DoD $120.0B · 3 Treasury $116.9B · 4 USPS $79.1B · 5 Education $48.4B	**PASS** 5/5 figures, order OK	Pending
14	Were any FY2024 agency net cost figures restated in the FY2025 report, and by how much? *intent: restatement*	Total $7,402.9B→$7,363.1B (–39.8) · Security Assist. $74.4B→$46.5B (–27.9) · DoD $1,232.3B→$1,218.4B (–13.9) · All other $19.3B→$23.2B (+3.9)	**PASS** 12/12 figures	Pending
15	Compare the net cost of HHS, SSA, and DoD between FY2024 and FY2025. *intent: comparison*	HHS +$148.1B (+8.5%) · SSA +$125.2B (+8.2%) · DoD +$5.0B (+0.4%)	**PASS** 12/12 figures	Pending

Notes. Q4: the tool returns the 10 lowest values; ranks 8–10 are positive, so the reasoner must report only the negative ones (all 7 expected agencies appear, TVA/FCSIC tie at –$0.6B). **Q9**: the ranking output carries net cost only; FDIC gross cost $21.9B and earned revenue $35.5B are not in the structured context (the answer would depend on vector chunks). **Q10**: restatement_differences correctly returns no differences, and the Total rows for both reports show $7,882.8B. **Q14**: all 12 expected figures including the four restated entities are returned.

7. Cross-cutting analysis and findings



#	Sev.	Finding	Evidence	Recommended fix
1	High	Fallback planner cannot read “FY2023”	`_heuristic_plan` uses `\b(20\d{2})\b`; in “FY2023” there is no word boundary between Y and 2, so fiscal_years is empty for all 15 questions (0%). It also ignores “Which 5 agencies” (only “top N”), picks *comparison* for “changes in assumptions” (Q2, Q7, Q12 → intent and metric wrong), misses “negative” as a ranking (Q4), and adds a spurious Treasury agency for “Interest on Treasury Securities” (Q8).	Use `(?:FY\s?)?(20\d{2})`; parse `\b(\d+)\s+agenc`; test “assumption” before “change”; add “negative” to ranking patterns.
2	Medium	Q9: ranking drops gross cost / earned revenue	In `retriever` the lookup call is skipped when intent is ranking even if an agency is named. Verified that lookup_financials(FDIC, 2024) returns $21.9B / $35.5B.	Run lookup_financials for the top-1..3 ranked agencies (or when agencies is non-empty).
3	Medium	Validator blind spots	(a) Small values pass: context contains list indices (“3.”) so an invented $3.0B matched; Q4’s +10% error was not caught. (b) The check is entity-blind: “VA had the highest net cost at $1,714.1B” passed because the figure exists somewhere in context. Large-value errors of ≥1% were caught.	Strip list indices / row numbers from the numeric pool; for ranking and lookup, additionally verify (entity, value) pairs from the structured rows.
4	Low	Ranking cannot express “negative only”	top_n must be guessed (Q4); non-negative rows are returned and must be filtered by the LLM.	Add max_value / only_negative parameter to rank_agencies.
5	Low	Ambiguous restatement message	“No restatement differences found (or no original + restated pair)” conflates ‘no change’ with ‘nothing to compare’ — risky for Q10-style yes/no answers.	Return two distinct messages; include counts of original/restated rows compared.

Strengths. Source-to-table ingestion is lossless across three formats; restatement handling is correct (originals preferred, restated rows preserved for Q10/Q14); all arithmetic (YoY deltas and percentages in Q3, Q5, Q8, Q15) is computed deterministically and matches ground truth to one decimal; guardrails block all probe attacks with no false blocks.

8. Completing the end-to-end (LLM) evaluation

From the project root, with a valid key in `.env` and the knowledge base built:

```
python ingest.py --reset
python evaluation/run_evaluation.py --mode live
```

This writes `evaluation/evaluation_results_live.json/.csv` with, per question, the final answer, validator verdict, figure recall in the answer, retries and latency. Suggested acceptance targets: answer figure recall ≥ 95%, validator pass ≥ 90% on first attempt, median latency ≤ 15 s. Record outcomes in the “Live LLM run” column above.

Files added to the project: `docs/Agentic_Workflow_Diagram.png`, `docs/Evaluation_Report.pdf`, `evaluation/run_evaluation.py`, `evaluation/evaluation_results_offline.json/.csv`. No existing file was modified.

Limitations. Offline retrieval used a hand-written reference plan, not LLM output; vector retrieval was unavailable, so recall is a lower bound for questions that also draw on Chroma chunks; the validator controls perturb only the first figure of each answer.